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SAMPLE POSITIONING DEVICE AND LIGHT SHEET MICROSCOPE

Abstract

A sample positioning device for a light sheet microscope includes a stage insert configured to be received by a stage of the light sheet microscope. The stage insert includes a rotatable sample holder. The rotatable sample holder includes a sample position configured to receive a sample carrier. The rotatable sample holder is configured to be rotatable around an axis of rotation that is parallel to an optical axis of the light sheet microscope, thereby enabling a rotation of a sample received by the sample carrier in a field of view of the light sheet microscope when the stage insert is received by the stage of the light sheet microscope.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims benefit to European Patent Application No. EP 24161436.1, filed on Mar. 5, 2024, which is hereby incorporated by reference herein.

FIELD

[0002] Embodiments of the present invention relate to a sample positioning device for a light sheet microscope, and to a light sheet microscope.

BACKGROUND

[0003] Light sheet microscopy is a fluorescence imaging technique. In light sheet microscopy, a sample is typically illuminated by a thin light sheet arranged perpendicular to the direction of observation. This configuration allows for optically sectioning a sample, enabling the selective illumination of specific planes within the sample. Multiple planes arranged next to each other in the sample can be sequentially illuminated and imaged, in order to image a volume of the sample. Thus, light sheet microscopy enables high-resolution, three-dimensional imaging with reduced phototoxicity and rapid data acquisition.

[0004] In order to acquire a volumetric image of the sample, the light sheet is moved through the sample along a scanning direction, for example using a galvo mirror. Typically, the acquired volumes are not cubic but stretched along the scanning direction of the light sheet. Depending on how the structure of interest is positioned in the sample, this internal orientation matches the scanning direction of the light sheet better or worse. Thus, in a typical light sheet microscope, the orientation of the imaged volumes is determined by the hardware. Consequently, to image a region of interest, it is often necessary to acquire and stitch together multiple volumes. Especially where speed is of the essence, for example with living samples, the time required to record multiple volumes offsets the time savings gained by using a light sheet illumination. With living samples in particular, there is also the problem that the sample can change between the images, so that artifacts can occur at the seams in a composite image.

SUMMARY

[0005] Embodiments of the present invention provide a sample positioning device for a light sheet microscope. The sample positioning device includes a stage insert configured to be received by a stage of the light sheet microscope. The stage insert includes a rotatable sample holder. The rotatable sample holder includes a sample position configured to receive a sample carrier. The rotatable sample holder is configured to be rotatable around an axis of rotation that is parallel to an optical axis of the light sheet microscope, thereby enabling a rotation of a sample received by the sample carrier in a field of view of the light sheet microscope when the stage insert is received by the stage of the light sheet microscope.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

[0007] FIG. 1 is a schematic view of a sample positioning device for a light sheet microscope according to an embodiment;

[0008] FIG. 2 is a schematic view of an exemplary light sheet microscope comprising a sample

positioning device according to an embodiment;

[0009] FIG. 3a is a schematic overview image in which a sample has not been aligned with a scanning direction according to some embodiments;

[0010] FIG. 3b is a schematic overview image in which individual regions in the sample have been align with the scanning direction according to some embodiments;

[0011] FIG. 4 a is a schematic view of a sample carrier before rotation and translation according to some embodiments; and

[0012] FIG. 4b is a schematic view of the sample carrier after rotation and translation according to some embodiments.

DETAILED DESCRIPTION

[0013] Embodiments of the present invention provide a sample positioning device for a light sheet microscope and a light sheet microscope which overcome the aforementioned disadvantages of the state of the art.

[0014] The sample positioning device for a light sheet microscope comprises a stage insert configured to be received by a stage of the light sheet microscope. The stage insert comprises a rotatable sample holder. The rotatable sample holder comprises a sample position configured to receive a sample carrier. The rotatable sample holder is configured to be rotatable around an axis of rotation that is parallel to an optical axis of the light sheet microscope thereby enabling a rotation of a sample received by the sample carrier in a field of view of the light sheet microscope when the stage insert is received by the stage of the light sheet microscope.

[0015] The rotatable sample holder enables the user to rotate the sample within the field of view of the light sheet microscope. This makes it possible to align the sample with the scanning direction of a light sheet generated by the light sheet microscope, and thus to align structures of interest in the sample with the volume imaged by the light sheet microscope. When the imaged volume is aligned with the structure of interest, the full recording speed can be utilized, so that there is only a minimal offset between the recorded planes in the imaged volume. At the same time the recording bandwidth is optimized, since the imaged volume contains everything relevant, but not much more than that. Another advantage is that fewer recorded volumes also mean significantly less exposure time and therefore less phototoxic stress on the sample. Thus, the proposed sample positioning device further improves on the two main advantages of light sheet microscopy, namely rapid three-dimensional imaging and reduced phototoxicity.

[0016] In an embodiment the stage insert is configured to be received in a recess of the stage, the recess being arranged along the optical axis of the light sheet microscope. The recess may be part of an opening of the stage which allows an observation of the sample from below and/or allows the sample to be illuminated from below. In particular, the recess may be formed by a ledge along the sides of such an opening. Such a ledge typically serves as a mounting point for accessories or additional equipment, such as slide holders or adaptors for non-standard sample containers. The recess provides support for the stage insert, and aids in its precise alignment and positioning.

[0017] Alternatively, the proposed sample positioning device for a light sheet microscope comprises a stage of the light sheet microscope. The stage comprises a rotatable sample holder and is configured to move the rotatable sample holder along at least one direction perpendicular to an optical axis of the light sheet microscope. The rotatable sample holder is arranged along the optical axis of the light sheet microscope, is configured to receive a sample carrier, and is configured to be rotatable around an axis of rotation that is parallel to an optical axis of the light sheet microscope to rotate a sample received by the sample carrier in a field of view of the light sheet microscope.

[0018] In this alternative embodiment, the rotatable sample holder is an integral part of the stage. Otherwise, the sample positioning device according to this alternative is identical to the sample positioning device comprising the stage insert described above. The sample positioning device comprising the stage has the same advantages as the sample positioning device comprising the stage insert. Both variants of the sample positioning device can be supplemented by the same

features described in this document.

[0019] In another embodiment, the sample positioning device comprises a drive unit configured to rotate the rotatable sample holder, and a control unit configured to control the drive unit. The drive unit may in particular comprise one or more electric motors, such as stepper motors or linear motors, or magnetic drives. In this embodiment, the rotation of the rotatable sample holder is motorized by the drive unit. Thereby, the rotation of the rotatable sample holder can be fully automated. A number of automatic and/or manual control schemes that may be performed by the control unit and/or the user in order to control the rotation of the rotatable sample holder are described below with reference to further embodiments.

[0020] In another embodiment, the control unit is configured to receive a user input and to control the drive unit to rotate the rotatable sample holder according to the received user input. The user input may be received, for example, via a user input device such as a keyboard and/or mouse. By the user input, a user directly or indirectly controls the drive unit, and thus the rotation of the rotatable sample holder. A number of exemplary user inputs and how they are used to control the rotation of the rotatable sample holder are described in the following with reference to further embodiments.

[0021] In another embodiment, the user input corresponds to a rotation specified by a user. The control unit is configured to control the drive unit to rotate the rotatable sample holder according to the user-specified rotation. The user-specified rotation may for example comprise a direction of rotation and/or an amount of angular displacement. In this embodiment, the user directly controls the rotation of the rotatable sample holder. In this embodiment in particular, the sample positioning device may comprise a dedicated user input device which allows the user to input the user-specified rotation. Such a dedicated user input device may comprise one or more rotatable knobs allowing the user to specify the rotation by rotating the knob. Since the rotation of the rotatable sample holder is automated, the user has a much finer control over the orientation of the sample than would be possible if the user manually oriented the sample, allowing the user to make very precise adjustments of the orientation.

[0022] In another embodiment, the user input corresponds to a region of interest in the sample, the region of interest being specified by the user. The control unit is configured to determine a rotation of the rotatable sample holder based on the user-specified region of interest and to control the drive unit to rotate the rotatable sample holder according to the determined rotation. In this embodiment, the user inputs a region of interest, for example by selecting the region of interest in an overview image of the sample displayed to the user. The region of interest may for example comprise a specific structure of the sample, such as an organ, or a single specimen in a collection of specimens. The control unit then determines the rotation of the rotatable sample holder, for example, such that the user-specified region of interest can be imaged with the minimum number of recordings, thereby reducing phototoxic stress on the sample, and maximizing recording bandwidth. Ideally, the control unit determines the rotation of the rotatable sample holder such that the user-specified region of interest aligns with the volume imaged by the light sheet microscope in a single recording. However, larger regions of interest may require imaging the region of interest in multiple recordings.

[0023] In another embodiment, the sample positioning device comprises an image processing unit configured to receive image data from the light sheet microscope and to determine a region of interest in the sample based on image data corresponding to an image of said sample captured by the light sheet microscope. In this embodiment, the image processing unit is configured to automatically determine the region of interest based in the image data. For example, the image processing unit may automatically detect structures in the sample and determine the region of interest based on the detected structures. In another example, the user provides as the user input which structures should be determined as the region of interest, and the image processing unit automatically detects the user-specified structure in the sample and determines the region of

interest accordingly. The user may in particular select the structure from a list of exemplary features displayed to the user, for example in the form of a drop-down menu. Orienting the sample such that the region of interest is aligned with the imaged volume has the above-mentioned advantages. In this embodiment, the determination of the region of interest is also automated, further speeding up image acquisition.

[0024] In another embodiment, the control unit is configured to control the drive unit to rotate the rotatable sample holder such that a scanning direction along which the light sheet generated by the light sheet microscope is scanned through the sample is aligned with the direction along which the region of interest has the largest extension. Typically, the distance over which the light sheet is moved along the scanning direction is larger than the extent of the focused area of the light sheet, i.e. the area of the light sheet which illuminates a single plane in the sample. Thus, by orienting the sample such that the direction along which the region of interest has the largest extension is aligned with the scanning direction, the number of recordings needed can be minimized. Minimizing the number of recordings has the above-mentioned advantages, in particular maximizing recording bandwidth by imaging as much of the region of interest as possible in a single recording.

[0025] In another embodiment, the sample positioning device comprises an image processing unit configured to receive image data from the light sheet microscope, to determine a rotation of the rotatable sample holder based on image data corresponding to an image of said sample captured by the light sheet microscope and to control the drive unit to rotate the rotatable sample holder according to the determined rotation. In this embodiment, the rotation is not necessarily determined based on a user-specified or automatically determined region of interest. Instead, the rotation may be determined based on other criteria, for example a geometric shape of structures in the sample which has been determined from the image data. Aligning the sample in that way has the above-mentioned advantages, in particular reducing phototoxic stress from multiple exposures.

[0026] In another embodiment, the control unit is configured to cause a translation of the stage in x- and/or y-direction. The control unit is in particular configured to cause the translation of the stage such that the field of view before a rotation of the rotatable sample holder has an overlap with the field of view after the rotation. For example, the translation of the stage and the rotation of the rotatable sample holder are combined such that the rotatable sample holder is rotated around the optical axis. In another example, the control unit may cause the translation of the stage such that the region of interest is moved into the field of view after a rotation of the rotatable sample holder. In both examples, it is ensured that an area of the sample that was observed before the rotation can still be seen in the field of view of the light sheet microscope after the rotation. This means that the user does not have to manually readjust the stage after a rotation in order to find the region of interest again.

[0027] In another embodiment, the sample positioning device comprises a control element configured to enable a user to manually rotate the rotatable sample holder. The control element may comprise knobs and screws, in particular a micrometer screw for fine adjustments. The control element may further comprise a handle-like element, similar to the handle of a crank, which allows the user to manually rotate the rotatable sample holder. Using the control element, the user can manually adjust the orientation of the rotatable sample holder, for example in order to align the sample with the scanning direction of the light sheet. Manual adjustment can be particularly faster when making changes over a large angular range, because the drive unit may operate with a very small step size.

[0028] Embodiments of the present invention further relate to a light sheet microscope, comprising a sample positioning device as described above. The light sheet microscope has the same advantages as the sample positioning device described above. In particular, the light sheet microscope may be supplemented with the features of the dependent claims directed at the sample positioning device. Furthermore, the sample positioning device described above may be supplemented with the features described in this document in connection with the light sheet

microscope.

[0029] In an embodiment the light sheet microscope comprises a single objective directed at the sample. In this embodiment, the light sheet microscope is configured as a single objective light sheet microscope, wherein the single objective is used for both illuminating the sample and receiving detection light from the sample, for example for the detection of a microscopic image of the sample. In such an arrangement, the light sheet is focused into the sample in such a way that it is oblique to the optical axis of the light sheet microscope. Because of this oblique position, such an optical arrangement is also called an oblique plane microscope. Since only a single objective is directed at the sample, the light sheet microscope can be designed to be particularly compact. It also allows imaging of samples that would be too large to be imaged with a typical light sheet microscopy setup involving two or more objectives directed at the sample.

[0030] FIG. 1 is a schematic view of a sample positioning device **100** for a light sheet microscope **200** according to an embodiment.

[0031] In order to generate volumetric images of a sample **202**, i.e. three-dimensional images of a sample **202**, a light sheet **204** generated by the light sheet microscope **200** is scanned through the sample **202** along a scanning direction C (c.f. FIG. 2). Thereby, a number of parallel planes inside the sample **202** can be sequentially illuminated and imaged. In a typical light sheet microscope **200**, the scanning direction C is fixed with respect to the hardware used for detection. The sample positioning device **100** allows a user to adjust the orientation of the sample **202**, for example to better align the sample **202** with the scanning direction C.

[0032] The sample positioning device **100** comprises a rotatable sample holder **102** which is exemplary formed as a turntable. The rotatable sample holder **102** is configured to be rotatable around an axis of rotation that extends vertically from the image plane of FIG. 1 and that is parallel to an optical axis O2 of the light sheet microscope **200**. A sample carrier **106**, exemplary shown as a Petri dish containing a number of biological specimens as the sample **202**, can be received by a sample position **106** formed by a top surface of the rotatable sample holder **102**.

[0033] Thus, a rotation of the rotatable sample holder **102** adjusts the orientation of the sample **202** inside the field of view of the light sheet microscope **200**. The rotation of the rotatable sample holder **102** is facilitated by a drive unit **108** of the sample positioning device **100**. The drive unit **108** is exemplary formed by a stepper motor engaged with a rim of the rotatable sample holder **102**.

[0034] In the embodiment shown in FIG. 1, the rotatable sample holder **102** is part of a stage insert **110**. The stage insert **110** is configured be arranged in an opening of a stage **112** of the light sheet microscope **200** which is covered from view by the stage insert **110** in FIG. 1. The opening is arranged such that the optical axis O2 of the light sheet microscope **200** passes through the opening. In other words, the opening is arranged along the optical axis O2 of the light sheet microscope **200**. The stage insert **110** exemplary fills the opening, and can be removed from the opening, for example to make room for other accessories, such as slide holders. A ledge **114** running around the inside of the opening mounts the stage insert **110** and prevents the stage insert **110** from falling through the opening. The ledge **114** is shown in FIG. 1 by a dotted line since it is covered from view by the stage insert **110**. In another embodiment, the rotatable sample holder **102** may be an integral part of the stage **112**. In such an embodiment, the stage **112** is a part of the sample positioning device **100**.

[0035] The stage **112** is configured to be moveable in two directions perpendicular to the optical axis O2 of the light sheet microscope **200** and the axis of rotation. In other words, the stage **112** is exemplary configured as an x-y-table. This allows the user to position the sample **202** within the field of view of the light sheet microscope **200**, for example in order to bring a region of interest into view. In the embodiment shown in FIG. 1, the stage **112** is motorized.

[0036] According to the embodiment shown in FIG. 1, the sample positioning device **100** also comprises a control unit **116**, exemplary formed by a desktop computer. The control unit **116** comprises input devices **118**, exemplary formed as mouse and keyboard, and is configured to

control the drive unit **108** and the stage **112**, in particular based on a user input received via at least one of the input devices **118**. This enables the user to control the position and orientation of the sample **202** using the sample positioning device **100** via a user input. In an example, the user specifies a rotation of the sample **202** by inputting an angle of rotation and a direction of rotation. [0037] The control unit **116** then controls the drive unit **108** to rotate the rotatable sample holder **102** by the user-specified amount in the user-specified direction. In another example, the user inputs a region of interest, for example by selecting the region of interest in an overview image of the sample **202** captured by the light sheet microscope **200**. The control unit **116** then controls the drive unit **108** to rotate the rotatable sample holder **102** such that the user-selected region of interest is best aligned with the light sheet **204** generated by the light sheet microscope **200**. This procedure is described in more detail below with reference to FIGS. **3a** and **3b**.

[0038] In this embodiment, the sample positioning device **100** further comprises an image processing unit **120** which is exemplary also formed as part of the control unit **116**. The image processing unit **120** is configured to process image data provided by the light sheet microscope **200**. In particular, the image processing unit **120** may be configured to determine a region of interest in an image captured by the light sheet microscope **200**. For example, the user may select a structure such as a specific organ or organelle from a list presented to the user on a display of the desktop computer. The image processing unit **120** then determines one or more regions of interest corresponding to the user-selected structure. The control unit **116** then controls the drive unit **108** to rotate the rotatable sample holder **102** such that the automatically determined region of interest is best aligned with the light sheet **204**. The image processing unit **120** may also directly determine the rotation of the rotatable sample holder **102** based on criteria other than a region of interest. In another example, the image processing unit **120** determines the rotation of the rotatable sample holder **102** such that a specimen is completely within the volume imaged by the light sheet microscope **200**.

[0039] FIG. **2** is a schematic view of an exemplary light sheet microscope **200** comprising a sample positioning device **100** according to an embodiment.

[0040] The light sheet microscope **200** further comprises an optical illumination system **206**, an optical relay system **208**, and an optical detection system **210**. The optical axis O3 of the optical detection system **210** is tilted relative to the optical axis O2 of the optical relay system **208** by an angle α . The optical axis O1 of the optical illumination system **206** is perpendicular to the optical axis O3 of the optical detection system **210**, so that the optical axis O1 of the optical illumination system **206** is tilted with respect to the optical axis O2 of the optical relay system **208** by an angle $90^\circ - \alpha$. The three aforementioned optical axes O1, O2, O3 intersect in an intermediate image space **212**.

[0041] The optical illumination system **206** comprises an illumination light source **214** and an illumination objective **216**. The illumination light source **214** generates the light sheet **204**, for example using a cylindrical lens not explicitly shown in FIG. **1**. Alternatively, the illumination light source **214** may generate the light sheet **204** as a quasi-static light sheet **204** by means of a dedicated scanning element. The light sheet **204** extends in a light propagation direction A and an expansion direction B perpendicular thereto. The two directions A, B are indicated by an oblique coordinate system shown in FIG. **1**. The illumination objective **216** images the light sheet **204** into the intermediate image space **212**.

[0042] The optical relay system **208** comprises an objective facing a sample **218**, a first tube lens **220**, a first ocular **222**, a second ocular **224**, a second tube lens **226** and a projection objective **228**, which are arranged in this order along the optical axis O2 of the optical relay system **208**. The optical relay system **208** images the light sheet **204** from the intermediate image space **212** into the sample **202**. Since the optical axis O1 of the optical illumination system **206** is tilted by $90^\circ - \alpha$ with respect to the optical axis O2 of the optical relay system **208**, the light propagation direction A is also tilted by $90^\circ - \alpha$ with respect to the optical axis O2 of the optical relay system **208**. Layers

within the sample **202** which are illuminated by the light sheet **204** are imaged back into the intermediate image space **212** by the optical relay system **208** as an intermediate image.

[0043] A scanning device **230** is arranged between the first ocular **222** and the second ocular **224**. The scanning device **230** is exemplarily formed by a movable mirror that deflects the optical axis O2 of the optical relay system **208**. The scanning device **230** is used to move the light sheet **204** through the sample **202** along the scanning direction C that is perpendicular to the optical axis O2 of the optical relay system **208** in the sample **202**. This allows the successive layers of the sample **202** along the scanning direction C to be illuminated so as to scan the volume of the sample **202**.

[0044] The optical detection system **210** comprises a detection objective **232**, a tube lens **234** and an image sensor **236**, arranged in this order starting from the intermediate image space **212**. The intermediate images of the illuminated layers in the intermediate image space **212** are imaged onto the image sensor **236** by the detection objective **232** and the tube lens **234**. In this way, layer images are generated on the image sensor **236**, each layer image corresponding to one of the layers in the sample **202** that are illuminated with the light sheet **204**. The layer images are captured by the image sensor **236** in form of raw image data and collected into an image stack.

[0045] The sample positioning device **100** allows the user to rotate the sample **202** arranged thereon around the optical axis O3 of the optical detection system **210**. Thereby, the sample's **202** orientation can be aligned with the scanning direction C. The extent of the focused part of the light sheet **204** along the light propagation direction A and the expansion direction B, i.e. the part of the light sheet **204** that is thin enough to be used for illumination, is much smaller than the extent to which the light sheet **204** can be moved along the scanning direction C through the sample **202** using the scanning device **230**. Thus, one advantageous arrangement is to rotate the sample **202** such that the scanning direction C is aligned with the direction along which the region of interest has the largest extension. This arrangement will be described in more detail below with reference to FIGS. **3a** and **3b**.

[0046] FIG. **3a** is a schematic overview image **300** in which the sample **202** is not aligned with the scanning direction C.

[0047] The sample **202** exemplarily comprises of a number of randomly oriented *C. elegans* specimens **302a**, **302b** which are nematodes having an elongated rod-like body. Two of the specimens **302a**, **302b** are to be imaged. FIG. **3a** also shows dashed rectangles **304a**, **304b**. Each of the rectangles **304a**, **304b** corresponds to a volume that can be imaged by the light sheet microscope **200** in a single recording, i.e. a single sweep of the light sheet **204** along the scanning direction C. In this example, the scanning direction C is right to left in FIG. **3a**. Thus, in order to image the two specimen **302a**, **302b**, six recordings are necessary. This increases phototoxic stress on the specimens **302a**, **302b** by exposing each specimen **302a**, **302b** three times to the illumination light. Further, since the specimen **302a**, **302b** are still alive, they may move between recordings. This results in artifacts at the seams in the composite volumetric image generated from the three separately captured volumes. The artifacts may in turn cause a loss of relevant information.

[0048] FIG. **3b** is a schematic overview image **300** in which individual regions in the sample **202** have been aligned with the scanning direction C.

[0049] Contrary to the situation depicted in FIG. **3a**, in FIG. **3b** there are only two recordings needed to image both specimens **302a**, **302b**. The volumes images in the situation depicted in FIG. **3b** are shown as dotted rectangles **306a**, **306b**. To image the first specimen **302a**, the sample **202** is rotated about 45° counterclockwise and moved laterally to the optical axis O2 such that the rectangle **306a** can be swept by the light sheet **204**. To image the second specimen **302b**, the sample **202** is then rotated about 80° clockwise and moved laterally to the optical axis O2 such that the rectangle **306b** can be swept by the light sheet **204**. This ensures, that the scanning direction C is aligned along the rod-shaped body of the specimens **302a**, **302b** in each recording, minimizing the number of recordings needed to image the whole specimen **302a**, **302b**. The rotation and movement of the sample **202** is described in more detail below with reference to FIGS. **4a** and **4b**.

[0050] In order to determine the afore-mentioned rotation, the user may first select one or more regions of interest, in this case the two specimens **302a**, **302b** of *C. elegans* that are to be imaged. The control unit **116** then determines the rotations from the user-selected regions of interest. Alternatively, the regions of interest may be determined automatically by the image processing unit **120** based on image data corresponding to the overview image **300** shown in FIGS. **3a** and **3b**. When a rotation would cause the region of interest to move out of the field of view of the light sheet microscope **200**, the control unit **116** may control the stage **112** to position the region of interest back into the field of view.

[0051] FIG. **4a** is a schematic view of the sample carrier **104** before rotation and translation.

[0052] FIG. **4a** shows the sample carrier **104** exemplary comprising the two specimens **302a**, **302b** of *C. elegans* as sample **202**. Further, FIG. **4a** shows the region of the sample over which the light sheet **204** can be swept in order to generate a volumetric image of the sample **202** as a dotted rectangle **400**, and the scanning direction C as an arrow. The area denoted by the rectangle **400** is fixed relative to the rest of the light sheet microscope **200**. Thus, in order to image the whole specimen **302a**, the sample **202** needs to be repositioned, i.e. rotated around the optical axis O2 using the rotatable sample holder **102** and moved laterally to the optical axis O2 by the stage **112**.

[0053] FIG. **4b** is a schematic view of the sample carrier **104** after rotation and translation.

[0054] As can be seen in FIG. **4b**, the sample **202** has been rotated by about 45° relative to its original position depicted in FIG. **4a**. The rotation is facilitated by the rotatable sample holder **102** and shown in FIG. **4b** by an arrow R. Additionally, the sample **202** has been moved to the right and slightly up relative to its original position depicted in FIG. **4a** using the stage **112**. This translation of the sample **202** is shown in FIG. **4b** by an arrow T. Using the rotatable sample holder **102** and the stage **112**, the sample **202** has been repositioned such that the elongated body of the specimen **302a** is fully within the area denoted by the rectangle **400**, i.e. the volume imaged by the light sheet microscope **200**.

[0055] Identical or similarly acting elements are designated with the same reference signs in all Figures. As used herein the term “and/or” includes any and all combinations of one or more of the associated listed items and may be abbreviated as “/”.

[0056] Although some aspects have been described in the context of an apparatus, it is clear that these aspects also represent a description of the corresponding method, where a block or device corresponds to a method step or a feature of a method step. Analogously, aspects described in the context of a method step also represent a description of a corresponding block or item or feature of a corresponding apparatus.

[0057] While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.

[0058] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity

from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

REFERENCE SIGNS

[0059] **100** Sample positioning device [0060] **102** Rotatable sample holder [0061] **104** Sample carrier [0062] **106** Sample position [0063] **108** Drive unit [0064] **110** Stage insert [0065] **112** Stage [0066] **114** Ledge [0067] **116** Control unit [0068] **118** Input device [0069] **120** Image processing unit [0070] **200** Light sheet microscope [0071] **202** Sample [0072] **204** Light sheet [0073] **206** Optical illumination system [0074] **208** Optical relay system [0075] **210** Optical detection system [0076] **212** Intermediate image space [0077] **214** Illumination light source [0078] **216** Illumination objective [0079] **218** Objective [0080] **220** Tube lens [0081] **222, 224** Ocular [0082] **226** Tube lens [0083] **228** Projection objective [0084] **230** Scanning device [0085] **232** Detection objective [0086] **234** Tube lens [0087] **236** Image sensor [0088] **300** Overview image [0089] **302a, 302b** Specimen [0090] **304a, 304b, 306a, 306b, 400** Rectangle [0091] A Propagation direction [0092] B Expansion direction [0093] C Scanning direction [0094] R, T arrow

Claims

1. A sample positioning device for a light sheet microscope, the sample positioning device comprising: a stage insert configured to be received by a stage of the light sheet microscope, the stage insert comprising a rotatable sample holder, wherein the rotatable sample holder comprises a sample position configured to receive a sample carrier, and wherein the rotatable sample holder is configured to be rotatable around an axis of rotation that is parallel to an optical axis of the light sheet microscope, thereby enabling a rotation of a sample received by the sample carrier in a field of view of the light sheet microscope when the stage insert is received by the stage of the light sheet microscope.
2. The sample positioning device according to claim 1, wherein the stage insert is configured to be received in a recess of the stage, the recess being arranged along the optical axis of the light sheet microscope.
3. A sample positioning device for a light sheet microscope, the sample positioning device comprising: a stage of the light sheet microscope, the stage comprising a rotatable sample holder, the stage being configured to move the rotatable sample holder along at least one direction perpendicular to an optical axis of the light sheet microscope, wherein the rotatable sample holder is arranged along the optical axis of the light sheet microscope, wherein the rotatable sample holder comprises a sample position configured to receive a sample carrier, and wherein the rotatable sample holder is configured to be rotatable around an axis of rotation that is parallel to the optical axis of the light sheet microscope to rotate a sample received by the sample carrier in a field of view of the light sheet microscope.
4. The sample positioning device according to claim 1, further comprising a drive unit configured to rotate the rotatable sample holder, and a control unit configured to control the drive unit.
5. The sample positioning device according to claim 4, wherein the control unit is configured to receive a user input, and to control the drive unit to rotate the rotatable sample holder according to the received user input.
6. The sample positioning device according to claim 5, wherein the user input corresponds to a rotation specified by a user, and the control unit is configured to control the drive unit to rotate the rotatable sample holder according to the user-specified rotation.
7. The sample positioning device according to claim 5, wherein the user input corresponds to a region of interest in the sample, the region of interest being specified by the user, and wherein the control unit is configured to determine a rotation of the rotatable sample holder based on the user-specified region of interest, and to control the drive unit to rotate the rotatable sample holder according to the determined rotation.

- 8.** The sample positioning device according to claim 4, further comprising an image processing unit configured to receive image data from the light sheet microscope, and to determine a region of interest in the sample based on the image data corresponding to an image of the sample captured by the light sheet microscope.
 - 9.** The sample positioning device according to claim 7, wherein the control unit is configured to control the drive unit to rotate the rotatable sample holder such that a scanning direction, along which a light sheet generated by the light sheet microscope is scanned through the sample, is aligned with a direction along which the region of interest has a largest extension.
 - 10.** The sample positioning device according to claim 4, further comprising an image processing unit configured to receive image data from the light sheet microscope, to determine a rotation of the rotatable sample holder based on the image data corresponding to an image of the sample captured by the light sheet microscope, and to control the drive unit to rotate the rotatable sample holder according to the determined rotation.
 - 11.** The sample positioning device according to claim 4, wherein the control unit is configured to cause a translation of the stage in x- and/or y-direction.
 - 12.** The sample positioning device according to claim 1, further comprising a control element configured to enable a user to manually rotate the rotatable sample holder.
 - 13.** A light sheet microscope, comprising the sample positioning device according to claim 1.
 - 14.** The light sheet microscope according to claim 13, wherein the light sheet microscope comprises a single objective directed at the sample.
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